Homework 5

Modules 5–6 · Homework · No date

Name _____ Date _____

1. A single 500mg dose of medication was administered orally. The medication is about 88% orally bioavailable and is characterized as having an absorption half-life of approximately 75 minutes, elimination half-life of approximately 4 hours, and apparent volume of distribution of 25 L.

1a Module 5

When does the maximum concentration of drug in the plasma occur?

1b Module 5

What is the expected maximum concentration following this single oral dose?

1c Module 5

What is the expected AUC following this single oral dose?

1d Module 5

If the dose were increased to 1000 mg, when does the maximum concentration of drug in the plasma occur?

1e Module 5

If the dose were increased to 1000 mg, what is the maximum concentration of drug in the plasma that you would expect?

1f Module 5

If the dose were increased to 1000 mg, what is the AUC that you would expect?

2. A 78 kg male patient received multiple IV bolus injections of 15 mg/kg every 8 hours for 48 hours. The medication has an elimination half-life of 4 hours and an apparent

volume of distribution that is 22% of body weight.

2a Module 6

What is the average concentration of drug in the plasma at steady state?

2b Module 6

What is the maximum concentration of drug in the plasma at steady state?

2c Module 6

What is the minimum concentration of drug in the plasma at steady state?

2d Module 6

What is the expected concentration of drug in the plasma 8 hours after administration of the last dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 3.05 hr

WHAT THE STEM GIVES

D_0	500mg	*not needed: t_{max} does not depend on the dose*
F	88%	*not needed: F scales height, not timing*
absorption t½	approximately 75 minutes	1.25 hours, so $k_a = \frac{0.693}{1.25} = 0.5544$ hr^{-1}
t½	approximately 4 hours	gives $k = \frac{0.693}{4} = 0.17325$ hr^{-1}
V_D	25 L	*not needed: t_{max} has no volume in it*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_a = \frac{0.693}{1.25\ hr} = 0.5544\ hr^{-1};\ k = \frac{0.693}{4\ hr} = 0.17325\ hr^{-1}$$

The absorption half-life of 75 minutes is 1.25 hours, so k_a, the absorption rate constant, is $\frac{0.693}{1.25\ hr} = 0.5544\ hr^{-1}$, and k, the elimination rate constant, is $\frac{0.693}{4\ hr} = 0.17325\ hr^{-1}$. Each is 0.693 over its own half-life because k t½ = ln 2; both are in reciprocal hours.

ALGEBRA

$$t_{max} = \frac{\ln(0.5544/0.17325)}{(0.5544 - 0.17325)\ hr^{-1}} = \frac{1.1632}{0.3811\ hr^{-1}} = 3.05\ hr$$

t_{max}, the time of the peak, is when the absorption rate equals the elimination rate: $t_{max} = \frac{\ln(k_a/k)}{k_a - k} \cdot \frac{0.5544}{0.17325} = 3.2$, a pure ratio of rate constants, and ln 3.2 = 1.1632; dividing by (k_a − k) = 0.3811 hr^{-1} inverts the reciprocal hours to hours, 3.05 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Absorption, half-life 75 minutes, is faster than elimination, half-life 4 hours, so the peak comes before one elimination half-life: 3.05 hr, between 1 and 4.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1b

Worked by the drill

Answer: 10.37 mg/L

WHAT THE STEM GIVES

D_0	500mg	in the numerator F k_a D_0
F	88%	written 0.88, in the numerator
absorption t½	approximately 75 minutes	gives $k_a = 0.5544\ hr^{-1}$, in the numerator and in $k_a - k$
t½	approximately 4 hours	gives $k = 0.17325\ hr^{-1}$, in $k_a - k$ and $e^{-k\ t_{max}}$
V_D	25 L	in the denominator $V_D(k_a - k)$

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_a = \frac{0.693}{1.25\ hr} = 0.5544\ hr^{-1};\ k = \frac{0.693}{4\ hr} = 0.17325\ hr^{-1}$$

The absorption half-life of 75 minutes is 1.25 hours, so k_a, the absorption rate constant, is $\frac{0.693}{1.25\ hr} = 0.5544\ hr^{-1}$, and k, the elimination rate constant, is $\frac{0.693}{4\ hr} = 0.17325\ hr^{-1}$. Each is 0.693 over its own half-life because k t½ = ln 2; both are in reciprocal hours.

ALGEBRA

$$C_{max} = \frac{(0.88)(500\ mg)(0.5544\ hr^{-1})}{(25\ L)(0.3811\ hr^{-1})}\ (e^{-(0.17325)(3.05)} - e^{-(0.5544)(3.05)})$$

The single oral dose equation $C_p = \frac{F\ k_a\ D_0}{V_D(k_a - k)}\ (e^{-kt} - e^{-k_a\ t})$ at $t = t_{max} = 3.05$ hr; F = 0.88 is the fraction absorbed, D_0 = 500 mg, V_D = 25 L. On top, (0.88)(500)(0.5544) = 243.94 mg·hr⁻¹; below, (25)(0.3811) = 9.529 L·hr⁻¹; the reciprocal hours cancel.

ROUND

$$C_{max} = (25.600)(0.4052) = 10.37\ mg/L$$

$\frac{243.94}{9.529} = 25.600$ mg/L is the coefficient. At t_{max} the elimination term $e^{-(0.17325)(3.05)} = 0.5895$ and the absorption term $e^{-(0.5544)(3.05)} = 0.1843$; their difference, 0.4052, times the coefficient gives C_{max} = 10.37 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
>
> If the whole absorbed dose were in the body at once the level would be $\frac{0.88 \times 500}{25}$, under 18 mg/L; some has been eliminated by 3.05 hr and a little is unabsorbed, so C_{max} is lower still: 10.37 mg/L.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1c

Worked by the drill

Answer: 101.6 (mg/L)·hr

WHAT THE STEM GIVES

D_0	500mg	in the numerator F D_0
F	88%	written 0.88, in the numerator

absorption t½	approximately 75 minutes	*not needed: AUC does not depend on k_a*
t½	approximately 4 hours	gives k = 0.17325 hr^{-1}, in the denominator k V_D
V_D	25 L	in the denominator k V_D

THE WORKING, ONE LINE AT A TIME

SETUP

$$AUC = \frac{F\ D_0}{k\ V_D} = \frac{(0.88)(500\ mg)}{(0.17325\ hr^{-1})(25\ L)} = 101.6\ (mg/L)\cdot hr$$

AUC, the area under the concentration–time curve, is the amount reaching the circulation over the clearance, because Cl × AUC = F D_0. Only the absorbed F = 0.88 of the 500 mg counts; Cl = k × V_D = (0.17325 hr^{-1})(25 L) = 4.3312 L/hr, and mg over L/hr is (mg/L)·hr.

> **DOES THE NUMBER MAKE SENSE?**
> AUC is the absorbed dose over clearance; 0.88 × 500 mg over (0.17325 hr^{-1})(25 L) comes to about 102 (mg/L)·hr: 101.6. Doubling the dose would double it.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1d

Worked by the drill

Answer: 3.05 hr

WHAT THE STEM GIVES

D_0	1000 mg	*not needed: t_{max} has no dose in it*
F	88%	*not needed*
absorption t½	approximately 75 minutes	gives k_a = 0.5544 hr^{-1}
t½	approximately 4 hours	gives k = 0.17325 hr^{-1}
V_D	25 L	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max} = \frac{\ln(k_a/k)}{k_a - k} = 3.05\ hr,\ unchanged$$

$t_{max} = \frac{\ln(k_a/k)}{k_a - k}$ contains only k_a = 0.5544 hr^{-1}, the absorption rate constant, and k = 0.17325 hr^{-1}, the elimination rate constant; the dose D_0 does not appear. Doubling the dose to 1000 mg changes neither constant, because both processes are first order, so the peak still comes at 3.05 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Both absorption and elimination are first order, so doubling the dose doubles every concentration but moves no timing: the peak is still at 3.05 hr, the part a value.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1e

Worked by the drill

Answer: 20.75 mg/L

WHAT THE STEM GIVES

D_0	1000 mg	twice 500 mg, so twice the part b peak
F	88%	unchanged
absorption t½	approximately 75 minutes	unchanged, so t_{max} is unchanged
t½	approximately 4 hours	unchanged
V_D	25 L	unchanged

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{max} = 2 \times 10.37 \text{ mg/L} = 20.75 \text{ mg/L}$

With first-order kinetics the whole curve scales with the dose, so twice the dose gives twice the peak at the same time.

> **DOES THE NUMBER MAKE SENSE?**
> C_{max} is proportional to the dose, so twice 500 mg gives twice 10.37 mg/L: 20.75 mg/L, above the part b peak.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1f

Worked by the drill

Answer: 203.2 (mg/L)·hr

WHAT THE STEM GIVES

D_0	1000 mg	twice 500 mg, so twice the part c area
F	88%	unchanged
absorption t½	approximately 75 minutes	*not needed: AUC does not depend on k_a*
t½	approximately 4 hours	unchanged
V_D	25 L	unchanged

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{AUC} = 2 \times 101.6 = 203.2 \text{ (mg/L)·hr}$$

AUC = $\frac{F\,D_0}{Cl}$: the amount absorbed over the clearance. F = 0.88 and Cl = k × V_D = 4.3312 L/hr do not change with the dose under first-order kinetics, so AUC is proportional to D_0; 1000 mg is twice 500 mg, and the area doubles from 101.6 to 203.2 (mg/L)·hr.

> **DOES THE NUMBER MAKE SENSE?**
> AUC is proportional to the dose at a fixed clearance, so twice 500 mg gives twice 101.6: 203.2 (mg/L)·hr, above the part c area.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2a

Worked by the drill

Answer: 49.19 mg/L

WHAT THE STEM GIVES

weight	78 kg	scales both the dose and the volume
D_0	15 mg/kg	scaled to 78 kg: 1170 mg, the numerator
τ	every 8 hours	in the denominator V_D k τ
duration	48 hours	*not needed for the number: 12 half-lives, so steady state is reached*
t½	4 hours	gives k = $\frac{0.693}{4}$ = 0.17325 hr^{-1}
V_D	22% of body weight	scaled to 78 kg: 17.16 L, in the denominator

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (15 \text{ mg/kg})(78 \text{ kg}) = 1170 \text{ mg};\ V_D = (0.22)(78 \text{ kg}) = 17.16 \text{ L};\ k = \frac{0.693}{4} = 0.17325 \text{ hr}^{-1}$$

The dose is 15 mg per kilogram and the volume of distribution V_D is 22% of body weight, so both are multiplied by the 78 kg weight: D_0 = 1170 mg and V_D = 17.16 L, the kilograms cancelling. k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{4 \text{ hr}}$ = 0.17325 hr^{-1}.

ALGEBRA

$$C_{avg}{}^{\infty} = \frac{D_0}{V_D\,k\,\tau} = \frac{1170 \text{ mg}}{(17.16 \text{ L})(0.17325 \text{ hr}^{-1})(8 \text{ hr})} = 49.19 \text{ mg/L}$$

$C_{avg}{}^{\infty}$, the average steady-state concentration, is the dose absorbed per interval over the plasma volume cleared per interval, Cl × τ with Cl = k × V_D: (17.16 L)(0.17325 hr^{-1})(8 hr) = 23.78 L, the hours cancelling. $\frac{1170 \text{ mg}}{23.78 \text{ L}}$ = 49.19 mg/L; F, the fraction absorbed, is 1 for an IV dose.

DOES THE NUMBER MAKE SENSE?

$C_0 = \frac{1170}{17.16}$ mg/L per dose and $k\tau = 0.17325 \times 8$, so the average is C_0 over $k\tau$, a little under three-quarters of C_0: 49.19 mg/L, between the steady-state trough and peak.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2b

Worked by the drill

Answer: 90.92 mg/L

WHAT THE STEM GIVES

weight	78 kg	scales both the dose and the volume
D_0	15 mg/kg	scaled to 78 kg: 1170 mg, the numerator of C_0
τ	every 8 hours	the exponent $k\tau = 0.17325 \times 8 = 1.386$
duration	48 hours	*not needed for the number: 12 half-lives, so steady state is reached*
t½	4 hours	gives $k = \frac{0.693}{4} = 0.17325\ \text{hr}^{-1}$
V_D	22% of body weight	scaled to 78 kg: 17.16 L, the denominator of C_0

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (15\ \text{mg/kg})(78\ \text{kg}) = 1170\ \text{mg};\ V_D = (0.22)(78\ \text{kg}) = 17.16\ \text{L};\ k = \frac{0.693}{4} = 0.17325\ \text{hr}^{-1}$$

The dose is 15 mg per kilogram and the volume of distribution V_D is 22% of body weight, so both are multiplied by the 78 kg weight: D_0 = 1170 mg and V_D = 17.16 L, the kilograms cancelling. k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{4\ \text{hr}} = 0.17325\ \text{hr}^{-1}$.

ALGEBRA

$$C_0 = \frac{1170\ \text{mg}}{17.16\ \text{L}} = 68.1818\ \text{mg/L};\ k\tau = (0.17325)(8) = 1.386;\ e^{-k\tau} = 0.2501$$

$C_0 = \frac{D_0}{V_D} = 68.1818$ mg/L is the rise in concentration each dose produces. $k\tau = (0.17325\ \text{hr}^{-1})(8\ \text{hr}) = 1.386$, a pure number, and $e^{-k\tau} = 0.2501$ is the fraction of each dose left when the next is given; the interval τ = 8 hr is 2 half-lives, so a quarter remains.

ALGEBRA

$$C_{max}^{\infty} = \frac{68.1818\ \text{mg/L}}{1 - 0.2501} = 90.92\ \text{mg/L}$$

Each dose adds C_0 = 68.1818 mg/L to what remains of the earlier doses; at steady state the peak is the sum $C_0(1 + e^{-k\tau} + e^{-2k\tau} + \ldots) = \frac{C_0}{1 - e^{-k\tau}}$. $1 - 0.2501 = 0.7499$, so $C_{max}^{\infty} = \frac{68.1818\ \text{mg/L}}{0.7499} = 90.92$ mg/L.

DOES THE NUMBER MAKE SENSE?

Eight hours is two half-lives of 4 hours, so a quarter of each dose remains at the next; the peak is C_0 = 68.1818 mg/L over 1 − 0.2501, a third more than 68.1818: 90.92 mg/L, between 68.1818 and 2 × 68.1818.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2c

Worked by the drill

Answer: 22.74 mg/L

WHAT THE STEM GIVES

weight	78 kg	scales the dose and the volume
D_0	15 mg/kg	scaled to 1170 mg; C_0 = 68.1818 mg/L, the numerator of the trough line
τ	every 8 hours	the exponent kτ, with $e^{-k\tau} = 0.2501$
duration	48 hours	*not needed for the number: steady state is reached*
t½	4 hours	gives k = 0.17325 hr^{-1}; 8 hours is two half-lives
V_D	22% of body weight	scaled to 17.16 L, the denominator of C_0

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (15\ \text{mg/kg})(78\ \text{kg}) = 1170\ \text{mg};\ V_D = 0.22 \times 78\ \text{kg} = 17.16\ \text{L};\ C_0 = \frac{1170\ \text{mg}}{17.16\ \text{L}} = 68.1818\ \text{mg/L}$$

The dose and the volume both scale with the 78 kg weight, so C_0, the rise in concentration each dose produces, is the dose over the volume.

SETUP

$$k = \frac{0.693}{4\ \text{hr}} = 0.17325\ \text{hr}^{-1};\ k\tau = 0.17325 \times 8 = 1.386;\ e^{-k\tau} = 0.2501$$

The printed trough line wants k and τ: k from the 4-hour half-life, and the 8-hour interval gives kτ, a pure number.

ALGEBRA

$$C_{min}^{\infty} = \frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}} = \frac{(68.1818\ \text{mg/L})(0.2501)}{1 - 0.2501} = 22.74\ \text{mg/L}$$

This is the printed trough line, read directly with C_0 from the stem: no peak is needed. Part b reaches the same number as the 90.92 mg/L peak times $e^{-k\tau}$.

DOES THE NUMBER MAKE SENSE?

Eight hours is two half-lives of 4 hours, so a quarter of each dose is left when the next goes in; the trough is C_0 times 0.2501 over 1 − 0.2501, a third of the 68.1818 mg/L C_0: near 22.74 mg/L, and below C_0.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2d

Worked by the drill

Answer: 22.74 mg/L

WHAT THE STEM GIVES

weight	78 kg	scales dose and volume inside the part b peak
D_0	15 mg/kg	inside C_{max}^{∞} = 90.92 mg/L from part b, the level the decay starts from
τ	every 8 hours	*not needed: the time asked is counted from the last dose*
duration	48 hours	*not needed for the number: steady state is reached*
t½	4 hours	gives k = 0.17325 hr^{-1} for the exponent kt = 0.17325 × 8
V_D	22% of body weight	inside the part b peak
t	8 hours after administration of the last dose	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$$C = C_{max}^{\infty}\, e^{-kt} = (90.92\ mg/L)\, e^{-(0.17325)(8)} = (90.92)(0.2501) = 22.74\ mg/L$$

After the last dose nothing more is given, so the level follows $C = C_0\, e^{-kt}$ from the steady-state peak C_{max}^{∞} = 90.92 mg/L, the concentration the moment the last dose is in. kt = (0.17325 hr^{-1})(8 hr) = 1.386, a pure number, and $e^{-1.386}$ = 0.2501 of the peak remains after 8 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is two half-lives of 4 hours, so a quarter of the 90.92 mg/L peak remains: 22.74 mg/L, the same as the trough, because the interval is also 8 hours.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.
